A Local Damage-Coupled Model for Ductile Fracture Under Combined Tension-Compression Loading

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Abstract

A local damage-coupled constitutive model is calibrated and validated against notched-bar and butterfly-specimen tests spanning stress states from uniaxial compression to plane-strain tension. The model reproduces the measured fracture strain locus across the full stress-state range tested, including the low-triaxiality compression-shear region where conventional uncoupled damage models are known to over-predict ductility.

1 Introduction

Combined tension-compression loading paths, typical of crash and impact events, subject a material to a stress-state history that most fracture calibration campaigns do not cover, since standard specimen geometries are tension-dominated. This paper extends a local damage-coupled model to the low- and negative-triaxiality regime using a purpose-built compression-shear specimen.

2 Material and Specimens

Three specimen geometries were used: smooth and notched round bars in tension, and a compression-shear butterfly specimen loaded on a modified Arcan fixture, giving stress triaxialities from -0.3 to 0.6.

3 Model and Calibration

The damage-coupled model follows a Lemaitre-type formulation with a stress-state-dependent damage evolution law,

$$\dot{D} = \left(\frac{Y}{S}\right)^{s} \dot{p}\, \eta(T^{*}), \tag{1}$$

where Y is the damage energy release rate, $\dot{p}$ the equivalent plastic strain rate, and $\eta(T^{*})$ a triaxiality-dependent weighting function calibrated against all three specimen geometries simultaneously.

4 Results and Validation

The calibrated model predicts fracture initiation location and applied displacement at fracture within 8% for a fourth, independent validation geometry (a flat-grooved plate) not used in calibration, across the full triaxiality range tested.

5 Conclusion

Coupling damage evolution to a triaxiality-dependent weighting function, calibrated jointly across tension- and compression-dominated specimens, removes the over-prediction of ductility in the low-triaxiality regime seen with uncoupled models calibrated on tension data alone.